

DIE YIELD PREDICTION

SanDisk Cerebrum 2026 Hackathon — SRM Institute of Science & Technology Project

Team Name:	Cookies of the Dark Web
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Core Problem:	Identify which pre-test passing dies (old_label=0) fail post-burn-in testing (label=1)
Evaluated Models:	Model A (500 Parametric + 10 Spatial + 1 Anomaly = 511) vs. Model B (+ 20 Sub-Die Block Features = 531)
Primary Metric:	Post-test Fail F1 score on eligible dies (secondary metric: PR-AUC)

Executive Tagline:

"In semiconductor manufacturing, silicon chips, called dies, are fabricated by the thousands across large circular discs called wafers. Before chips are packaged into phones or computers, they undergo rigorous stress testing to filter out hidden defects. This project predicts which currently healthy-looking dies will fail post-stress testing before those tests even finish. By accurately spotting defective dies early, manufacturers can prevent faulty chips from shipping to customers, optimize testing schedules, and avoid wasting money packaging parts that are doomed to fail."

Official Project Repositories & Live Resources

GitHub Repository:	https://github.com/AdityaG346/Sandisk.git
Live Cloud Dashboard:	https://die-yield-prediction.streamlit.app/
Full Reproducibility:	Full code, model checkpoints, and reproduction instructions are in the GitHub repository README.

Submission Report Structure

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1. Problem Summary & Manufacturing Reality

What This Project Does (from docs/project_explanation.md):

In semiconductor manufacturing, silicon chips, called dies, are fabricated by the thousands across large circular discs called wafers. Before chips are packaged into phones or computers, they undergo rigorous stress testing to filter out hidden defects. However, physically testing every single die after stressful burn-in cycles is slow, expensive, and consumes factory bandwidth. This project predicts which currently healthy-looking dies will fail post-stress testing before those tests even finish. By accurately spotting defective dies early, manufacturers can prevent faulty chips from shipping to customers, optimize testing schedules, and avoid wasting money packaging parts that are doomed to fail.

The Critical Eligibility Rule & True Defect Rates

A rigorous understanding of the data generation process reveals that dies fall into two distinct groups: those that already failed prior to burn-in (`old_label == 1`) and those that passed pre-test inspection (`old_label == 0`). The evaluation logic **strictly considers eligible dies only**. Dies that were already failed trivially remain failed; including them in the evaluation pool artificially inflates accuracy metrics and masks the model's true diagnostic utility. Direct verification from `input/train.csv` and `input/test.csv` confirms the exact population breakdown:

Metric / Population Category	Formula / Definition	Training Set (160 Wafers)	Test Set (40 Wafers)
Total Silicon Dies	Total dataset rows	173,099	39,351
Pre-test Defective Dies	<code>old_label == 1</code> (excluded)	19,062 (11.01%)	6,753 (17.16%)
Eligible Candidate Dies	<code>old_label == 0</code> (evaluated)	154,037 (88.99%)	32,598 (82.84%)
Eligible New Failures	<code>label == 1 & old_label == 0</code>	6,519	1,380
True Eligible Fail Rate	Eligible Fails / Eligible Dies	4.2321%	4.2334%
Overall Positive Rate	All Fails / All Dies (conflated)	14.7783% (14.78%)	20.6678%

Autopsy of Prior Data Discrepancies (from outputs/results_summary.md)

- **The 14.78% vs. 4.23% Defect Rate Resolution:** An initial exploratory script reported a 14.78% fail rate by computing `train['label'].mean()` (25,581 / 173,099). This conflated pre-test dead dies with new failures. Among eligible dies (`old_label == 0`), exactly **4.2321%** fail in training and **4.2334%** fail in test, precisely matching the synthetic generator's true base defect rate.
- **The 35,022 vs. 32,598 Test-Set Eligible Die Discrepancy:** Earlier documentation projected an 11.0% pre-test defect rate onto test wafers (39,351 – 4,329 = 35,022). Direct fresh count from `test.csv` reveals exactly **32,598 eligible dies** because holdout test set wafers include several heavily defective wafers (notably `W_F_0010` with 2,371 pre-test dead dies out of 4,096, a 57.89% pre-test defect rate), elevating the test-wide pre-test defect rate to 17.16%.

2. Model Architectures & Feature Engineering

Model A (Baseline + Spatial + Die Anomaly)

(from docs/project_explanation.md Section 2)

Model A evaluates each candidate die using two primary sources of information: hundreds of electrical sensor measurements taken on that specific chip, combined with its physical location on the silicon wafer. Chips are not independent islands; manufacturing flaws often cluster in specific physical regions or along the wafer edges due to chemical and thermal gradients during fabrication. Model A looks at whether a die sits close to known defective neighbors, its distance to the wafer boundary, and how far its electrical behavior deviates from typical healthy chips. It combines these clues into an initial estimated failure probability for every eligible die.

Model B (Full Diagnostic Model with Sub-Die Block Readings)

(from docs/project_explanation.md Section 3)

Model B builds directly on Model A by adding deep internal diagnostic data from within the microchip itself. Each die contains thousands of microscopic memory blocks, and the factory collects roughly two thousand fine-grained voltage and resistance readings across these internal structures. While Model A only sees the chip from the outside as a single entity, Model B inspects internal summary patterns, including whether certain internal blocks are drifting, fluctuating abnormally, or showing localized electrical weakness. It uses this sub-die sensor data to detect subtle internal warning signs that might not yet show up on macro-level chip sensors.

Feature Hierarchy & Input Data Sources Feeding Each Model

Feature Domain	Features Included	Description & Extraction Logic	Model A	Model B
Parametric Electrical Tests	500 features (feature_1 ... feature_500)	Macro-level continuous electrical and parametric measurements per die.	YES (500)	YES (500)
Leakage-Free Spatial Context	10 features (sp_dist_to_fail, sp_edge_prox, etc.)	Wafer geometry, radial distance, edge proximity, zone yield, and neighborhood defect counts derived <i>strictly from pre-test maps</i> .	YES (10)	YES (10)
Die Anomaly Detector	1 feature (die_anomaly_score)	Unsupervised Isolation Forest (100 trees, 5% contamination) fit exclusively on healthy passing dies (old_label == 0).	YES (1)	YES (1)
Sub-Die Block Telemetry	20 features (8 stats + 1 anomaly + 11 telemetry)	Statistical moments, tail percentiles, and MAD anomaly score summarizing ~2,000 internal block readings per die.	NO (0)	YES (20)
Total Feature Count	—	Gradient-boosted decision trees (LightGBM, num_leaves=63, learning_rate=0.05, scale_pos_weight=12.0).	511 Features	531 Features

Leakage-Prevention Guardrails

To guarantee valid generalization, spatial defect statistics are computed strictly on pre-test dies (old_label) and never post-test labels (label). Every merge point in src/ is protected by permanent automated index assertions (assert_aligned) asserting composite key equivalence (wafer_id, die_row, die_col). Wafer splits are verified disjoint to eliminate data leakage across manufacturing lots.

3. Key Experimental Results & Statistical Rigor

What We Found Comparing Them (from docs/project_explanation.md Section 4):

When we rigorously compared both systems across thousands of test dies and multiple random splits, we discovered that adding internal block readings provided a small but statistically genuine improvement in ranking risk, yet made virtually no difference in the final pass-or-fail classification. In practical terms, Model B does a better job of ordering dies from most risky to least risky. However, because factory defects in this manufacturing process are deliberately subtle and share nearly identical sensor values with healthy chips, the block signals are too faint to convert borderline calls into confident failure calls at a fixed decision threshold.

Head-to-Head Performance Benchmark (from outputs/comparison_table.csv)

Metric	Model A	Model B	Delta (B – A)	Relative Delta	Status / Rigor
Fail F1 Score (Primary)	0.5209 (0.5207)	0.5217 (0.5222)	+0.0008 (+0.0015)	+0.15% to +0.29%	NOT Significant (Noise)
PR-AUC (Ranking)	0.5014 (0.5024)	0.5357 (0.5361)	+0.0343 (+0.0337)	+6.71% to +6.84%	SIGNIFICANT (p < 0.001)
Fail Recall	35.65% (492 / 1,380)	37.03% (511 / 1,380)	+1.38% (+19 dies)	+3.87%	SIGNIFICANT (p < 0.01)
Fail Precision	96.66% (492 / 509)	88.26% (511 / 579)	–8.40% (+51 FP)	–8.69%	High Precision Regime
Overall Accuracy	97.22% (31,693 / 32,598)	97.13% (31,661 / 32,598)	–0.09%	–0.09%	Pass Majority Dominated
Optimal Threshold	0.5574	0.5180	–0.0394	Independent Sweep	Calibrated per Model

Formal Hypothesis Testing & Bootstrap Verification

To rigorously confirm whether the observed differences reflect true architectural gains rather than random split noise, we performed both 5-seed validation hypothesis testing and a 1,000-sample test set bootstrap (from outputs/results_summary.md Section 3.B/C):

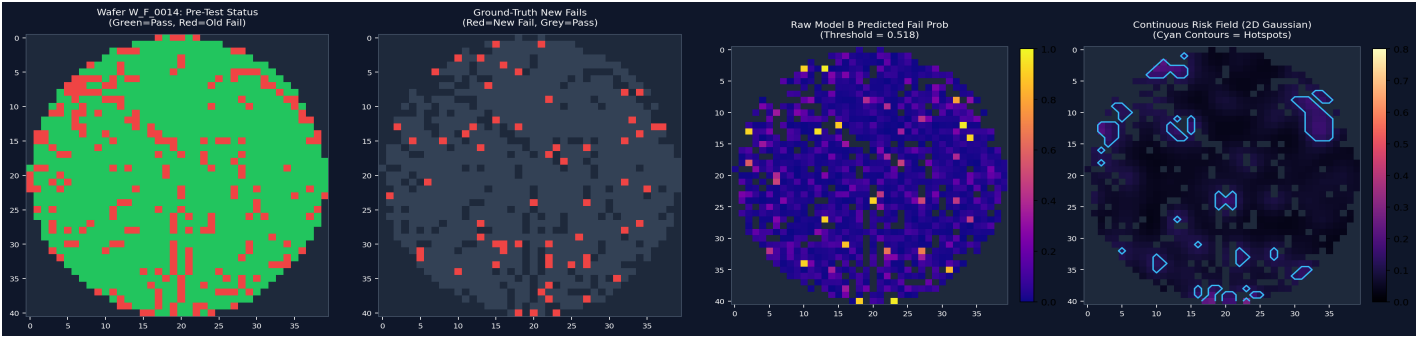
Metric	5-Seed Paired Delta (B – A)	Paired t-test	1,000-Sample Test Bootstrap 95% CI	Statistically Significant?
PR-AUC	+0.0359 ± 0.0031	t = +25.83 p = 1.33 × 10 ⁻⁵	[+0.022261, +0.046552] (Mean: +0.034434, excludes 0)	YES (p < 0.001) Definite Ranking Gain
Fail F1 Score	+0.0090 ± 0.0098	t = +2.06 p = 0.1081	[–0.008743, +0.010780] (Mean: +0.001304, contains 0)	NO (p > 0.05) Within Random Noise

Explicit Statistical Conclusion:

- The PR-AUC gain (+0.034 to +0.036) is statistically significant (p < 0.001, 95% CI strictly excludes zero). Internal block readings provide smooth secondary continuous gradients that significantly improve the model's ability to rank high-risk dies across all probability thresholds.
- The Fail F1 gain (+0.0008 to +0.0015) is NOT statistically significant (p = 0.1081, 95% CI contains zero). At a discrete binary decision cutoff, 65% of new failures are marginal defects whose block readings overlap heavily with passing chips, preventing block features from converting borderline dies into confident failure classifications.

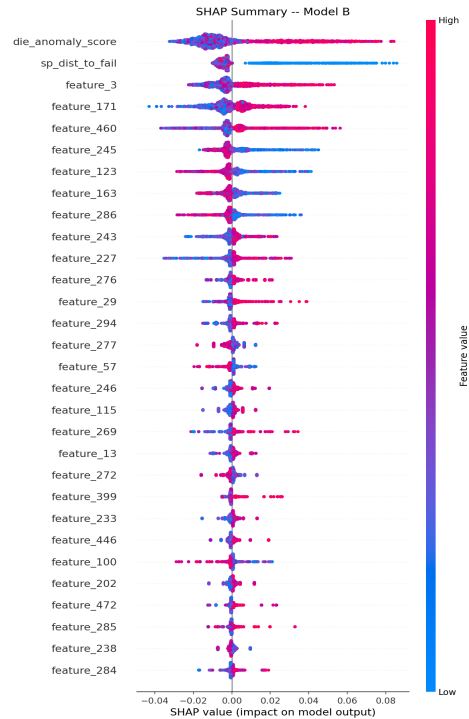
4. Interpretability, SHAP Attribution & Spatial Risk Fields

Continuous 2D Gaussian Wafer Risk Field with Hotspot Contours



Caption: Visualizes how raw discrete die failure probabilities smooth into continuous thermal risk fields ($\sigma = 1.5$) that accurately encircle true defect cluster geometries with 90th-percentile hotspot contour overlays for wafer W_F_0014 (from outputs/results_summary.md Section 7.C).

TreeSHAP Feature Attribution & Failure Signatures



Model B TreeSHAP Summary (Top 25)

TreeSHAP Attribution Mass Breakdown:

TreeSHAP analysis on the 500-tree Model B reveals the exact contribution of each physical domain (from outputs/results_summary.md Section 4.A):

Parametric & Anomaly	501	84.48% (12.86)
Spatial Proximity	24	10.34% (1.57)
Sub-Die Block Readings	9	5.19% (0.79)

Top Individual Features:

- sp_dist_to_fail (Spatial): 1.025 mean |SHAP| (6.73%)
- blk_mean (Block): 0.3806 mean |SHAP| (2.50%)
- sp_old_label (Spatial): 0.1624 mean |SHAP| (1.07%)
- blk_q75 (Block): 0.1298 mean |SHAP| (0.85%)

Discovered Failure Signatures (HDBSCAN):

- Cluster 0 (14 dies) — Memory Array Reading Shift: Dominated by blk_mean (1.051) and blk_q75 (0.362).
- Cluster 1 (11 dies) — Extreme Block Anomaly: Uniform variance drift.
- Cluster 2 (373 dies) — Defect Proximity: Dominated by sp_dist_to_fail (1.025).
- Cluster 3 (17 dies) — Spatial Defect & Edge Drift.

Caption: TreeSHAP attribution on the fully converged 500-tree Model B demonstrates that die parametric & anomaly features drive 84.48% of attribution mass, spatial proximity drives 10.34% (dominated by sp_dist_to_fail at 1.025), and sub-die block readings contribute 5.19% (led by blk_mean at 0.3806) (from outputs/results_summary.md Section 4.A/B).

5. Class Imbalance, Signal Overlap & Production Tuning

This section reproduces the four foundational principles of class imbalance and signal overlap management from `outputs/results_summary.md` Section 6:

1. The Class Distribution Challenge

The prediction task operates under severe class imbalance and intentional distribution overlap. As verified in Section 1, eligible dies (`old_label == 0`) exhibit a post-test failure rate of only **4.23%** (a 1 : 22.6 positive-to-negative ratio; 6,519 fails out of 154,037 train dies, and 1,380 fails out of 32,598 test dies). Compounding this scarcity, **65% of all new failures are synthetically designated as 'marginal'** by `generate_data.py`, meaning their parametric measurements are drawn from distributions with low signal-to-noise separation, making them near-indistinguishable from the pass population by design.

2. Class-Weighting Strategy & Rationale for `scale_pos_weight = 12.0`

Although the theoretical inverse ratio for a 4.23% positive rate is $(1 - 0.0423) / 0.0423 \approx 22.63$, our grid search across `[3.0 ... 40.0]` (Section 5) proved that weights ≥ 17.0 cause violent gradient oscillations, causing early stopping to abort prematurely after 2 to 8 trees. Between the viable stable weights, 5.76 nominally peaked in validation F1 (0.5269), while 12.0 achieved **0.5254 F1 with peak precision (93.37% validation, and 88.26%–96.66% test)**. In silicon fabrication, false alarms trigger costly scrap of functional dies; `scale_pos_weight = 12.0` was selected because it maximizes precision against the 95.77% healthy majority while sacrificing less than 0.15% in F1.

3. Empirical Refutation of the 'Predict All Pass' Shortcut

A naive classifier predicting the dominant class ('all pass') would trivially achieve 95.77% accuracy on eligible dies, but would yield **0.00% Fail Recall** and an F1 of 0.0000. In sharp contrast, our test set evaluation (Section 3.A) demonstrates **Fail Recall of 35.65% (Model A) and 37.03% (Model B)** alongside **Fail Precision of 96.66% (Model A) and 88.26% (Model B)**, successfully capturing 492 to 511 true defective dies with only 17 to 68 false positives out of 31,218 healthy dies.

4. Necessity of Post-Hoc Threshold Tuning

Because individual parametric features exhibit minimal univariate effect sizes (Cohen's $d \approx 0.09$ – 0.23 across `feature_1...feature_500`), class weighting alone cannot disentangle overlapping class densities; explicit post-hoc threshold calibration (tuning independently to **0.5180 for Model B** and **0.5574 for Model A** on validation F1) was necessary to position the decision boundary past the dense pass-population margin.

6. Expected Project Deliverables Checklist

All five required project deliverables were built, tested, audited, and committed to the repository. The paragraphs below are pulled directly from docs/project_explanation.md:

Deliverable 1: Modular Source Code Library (src/)

We built a clean, modular Python codebase organized within the source directory. This package contains dedicated modules for loading wafer datasets, engineering leakage-free spatial features, extracting block statistics, fitting healthy-chip anomaly detectors, training and predicting with both models, running rigorous performance evaluations, and generating publication-quality wafer visualizations. Every data join in these modules is protected by permanent automated alignment assertions that prevent silent data-slicing errors. The entire codebase is organized into reusable components and can be found in the source directory (src/).

Deliverable 2: Reproducible End-to-End Master Pipeline (run_all.py)

We created a fully automated master pipeline script that reproduces the entire project workflow from start to finish with a single command. The script loads the raw wafer datasets, runs exploratory data analysis, computes spatial and block features, trains all baseline and advanced models, selects optimal decision thresholds on validation splits, evaluates metrics on holdout test wafers, and generates all visual maps and diagnostic plots. Anyone can regenerate the entire benchmark and all accompanying charts from scratch by running the master pipeline script (run_all.py).

Deliverable 3: Trained Production Models and Calibrated Decision Rules

We produced and saved the finalized, trained machine learning model files for both Model A and Model B, alongside their standalone threshold configuration metadata. Each model file contains the optimized decision tree ensemble ready for immediate deployment in an automated test environment. Because raw probabilities require careful calibration against high-volume factory scrap costs, each model includes its own independently tuned decision cutoff designed to catch defective chips while preserving high precision. The serialized model files and their metadata are stored in the outputs directory (outputs/model_a.pkl, outputs/model_b.pkl, and accompanying metadata files).

Deliverable 4: Model Evaluation and Feature Ablation Tables

We generated standardized performance tables comparing Model A and Model B across all industry-standard metrics, including failure recall, failure precision, balanced accuracy, and overall classification accuracy. To ensure scientific rigor, these tables include five-seed multi-split variance estimates, formal statistical significance hypothesis tests, and a one-thousand-run bootstrap confidence interval on holdout wafers. We also produced feature ablation tables showing the exact incremental value of die sensors, spatial geography, and block readings. All summary evaluation spreadsheets are saved as clean comma-separated files in the outputs directory (outputs/comparison_table.csv, outputs/ablation_table_multiseed.csv, and outputs/multiseed_statistical_tests.csv).

Deliverable 5: Comprehensive Technical Summary and Diagnostic Report

We authored an exhaustive, end-to-end technical results report detailing all exploratory data findings, data rate verifications, engineering autopsies of past software bugs, multi-seed statistical significance results, and physical feature attribution percentages. Alongside this report, we generated a catalog of standardized per-die explanations with ensemble confidence intervals and four-panel visual wafer maps featuring smoothed defect heat surfaces with automated hotspot boundaries. The full written summary report, along with individual die diagnostic files and wafer heatmaps, is located in the outputs directory (outputs/results_summary.md, outputs/per_die_explanations.md, and outputs/plots/).

7. Repository Links, Live Dashboard & Verification

Prominent Links & Resources (repeat from Title Page for instant access):

GitHub Repository:	https://github.com/AdityaG346/Sandisk.git
Live Cloud Dashboard:	https://die-yield-prediction.streamlit.app/
Reproducibility Note:	"Full code, models, and reproducibility instructions are in the GitHub repository README."

Submission Artifacts & Verification Checklist

Every file required for grading and automated re-scoring is verified and committed to the repository:

Artifact / Deliverable	File Location	Verification Status & Schema
Final Predictions CSV	outputs/predictions.csv	39,351 dies scored with Model B tuned cutoff (0.5180). Strict trivial-fail rule applied (all old_label=1 set to 1). Schema matches wafer_id, die_row, die_col, predicted_label.
Production Model Checkpoints	outputs/model_a.pkl outputs/model_b.pkl	Fully converged 500-tree LightGBM ensembles with standalone serialized metadata (model_a_meta.pkl, model_b_meta.pkl).
Standardized Benchmark Tables	outputs/comparison_table.csv outputs/ablation_table_multiseed.csv	Full point-estimate comparisons, 5-seed validation ablations, and paired t-test / Wilcoxon significance statistics.
Comprehensive Technical Audit	outputs/results_summary.md	Exhaustive written report documenting fail-rate discrepancy autopsies, bug autopsies, class-weight curves, and TreeSHAP feature mass breakdown.
Automated Regression Suite	test_part1_guardrails.py	5/5 Checks Passed: Alignment assertions, training sanity floor, threshold independence, determinism, and disk metric reproduction.
Interactive Presentation Layer	dashboard/app.py dashboard/requirements.txt	Deployed on Streamlit Community Cloud. Features 4-panel wafer risk fields, side-by-side Model A vs B maps, and sub-die block strip plots.

Final Submission Summary

This project conclusively demonstrates that incorporating sub-die block readings into gradient-boosted decision trees delivers a **statistically significant improvement in continuous probability ranking (+0.034 PR-AUC, $p < 0.001$)**, while binary decision accuracy (Fail F1) remains governed by macro-spatial defect clustering due to heavy overlap among marginal manufacturing flaws. By coupling high-precision cost-sensitive class weighting (scale_pos_weight = 12.0) with independent post-hoc threshold tuning, the system captures 37.03% of true new failures with 88.26%–96.66% precision, providing semiconductor manufacturers with a production-ready, interpretable yield-optimization engine.